

Critical (P_5, dart) -free graphsWen Xia^{a,b}, Jorik Jooken^c, Jan Goedgebeur^{c,d}, Shenwei Huang^{e,*}^a College of Computer Science, Nankai University, Tianjin 300071, China^b Tianjin Key Laboratory of Network and Data Security Technology, Nankai University, Tianjin 300071, China^c Department of Computer Science, KU Leuven Campus Kulak-Kortrijk, 8500 Kortrijk, Belgium^d Department of Applied Mathematics, Computer Science and Statistics, Ghent University, 9000 Ghent, Belgium^e School of Mathematical Sciences and LPMC, Nankai University, Tianjin 300071, China

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ABSTRACT

Given two graphs H_1 and H_2 , a graph is (H_1, H_2) -free if it contains no induced subgraph isomorphic to H_1 nor H_2 . A dart is the graph obtained from a diamond by adding a new vertex and making it adjacent to exactly one vertex with degree 3 in the diamond.

In this paper, we show that there are finitely many k -vertex-critical (P_5, dart) -free graphs for $k \geq 1$. To prove these results, we use induction on k and perform a careful structural analysis via the Strong Perfect Graph Theorem combined with the pigeonhole principle based on the properties of vertex-critical graphs. Moreover, for $k \in \{5, 6, 7\}$ we characterize all k -vertex-critical (P_5, dart) -free graphs using a computer generation algorithm. Our results imply the existence of a polynomial-time certifying algorithm to decide the k -colorability of (P_5, dart) -free graphs for $k \geq 1$ where the certificate is either a k -coloring or a $(k + 1)$ -vertex-critical induced subgraph.

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1. Introduction

All graphs in this paper are finite and simple. A k -coloring of a graph G is a function $\phi : V(G) \rightarrow \{1, 2, \dots, k\}$ such that $\phi(u) \neq \phi(v)$ whenever $uv \in E(G)$. Equivalently, a k -coloring of G can be viewed as a partition of $V(G)$ into k stable sets. If a k -coloring exists, we say that G is k -colorable. The chromatic number of G , denoted by $\chi(G)$, is the minimum number k such that G is k -colorable. A graph G is k -chromatic if $\chi(G) = k$. A graph G is k -critical if it is k -chromatic and $\chi(G - e) < \chi(G)$ for any edge $e \in E(G)$. For instance, K_2 is the only 2-critical graph and odd cycles are the only 3-critical graphs. A graph is critical if it is k -critical for some integer $k \geq 1$. Vertex-criticality is a weaker notion. A graph G is k -vertex-critical if $\chi(G) = k$ and $\chi(G - v) < k$ for any $v \in V(G)$.

For a fixed $k \geq 3$, it has long been known that determining the k -colorability of a general graph is an NP-complete problem [22]. However, the situation changes if one restricts the structure of the graphs under consideration.

Let $\mathcal{H}$ be a set of graphs. A graph G is $\mathcal{H}$ -free if it does not contain any member in $\mathcal{H}$ as an induced subgraph. When $\mathcal{H}$ consists of a single graph H or two graphs H_1 and H_2 , we write H -free and (H_1, H_2) -free instead of $\{H\}$ -free and $\{H_1, H_2\}$ -free, respectively. We say that G is k -vertex-critical $\mathcal{H}$ -free if it is k -vertex-critical and $\mathcal{H}$ -free. In this paper, we study k -vertex-critical $\mathcal{H}$ -free graphs. The following problem arouses our interest: Given a set $\mathcal{H}$ of graphs and an integer $k \geq 1$, are there finitely many k -vertex-critical $\mathcal{H}$ -free graphs? This question is very important because the finiteness of the set has a fundamental algorithmic implication.

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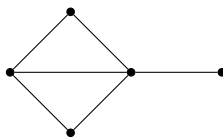


Fig. 1. The dart graph.

Theorem 1 (Folklore). *If the set of all k -vertex-critical $\mathcal{H}$ -free graphs is finite, then there is a polynomial-time algorithm to determine whether an $\mathcal{H}$ -free graph is $(k - 1)$ -colorable. $\square$*

Let K_n be the complete graph on n vertices. Let P_t and C_t denote the path and the cycle on t vertices, respectively. The complement of G is denoted by $\bar{G}$. For two graphs G and H , we use $G + H$ to denote the disjoint union of G and H . For a positive integer, we use rG to denote the disjoint union of r copies of G . For $s, r \geq 1$, let $K_{r,s}$ be the complete bipartite graph with one part of size r and the other part of size s . Our research is mainly motivated by the following two theorems.

Theorem 2 ([16]). *For any fixed integer $k \geq 5$, there are infinitely many k -vertex-critical P_5 -free graphs.*

It is natural to consider which subclasses of P_5 -free graphs have finitely many k -vertex-critical graphs. In 2021, Cameron, Goedgebeur, Huang and Shi [5] obtained the following dichotomy result.

Theorem 3 ([5]). *Let H be a graph of order 4 and $k \geq 5$ be a fixed integer. Then there are infinitely many k -vertex-critical (P_5, H) -free graphs if and only if H is $2P_2$ or $P_1 + K_3$.*

This theorem completely solves the finiteness problem of k -vertex-critical (P_5, H) -free graphs for $|H| = 4$. In [5], the authors also posed the natural question of which five-vertex graphs H lead to finitely many k -vertex-critical (P_5, H) -free graphs.

It is known that there are exactly 13 5-vertex-critical (P_5, C_5) -free graphs [16]. Recently, Cameron and Hoàng constructed infinite families of k -vertex-critical (P_5, C_5) -free graphs for $k \geq 6$ [7]. It has been proven that there are finitely many k -vertex-critical (P_5, banner) -free graphs for $k = 5$ [18] and 6 [4] and there are finitely many k -vertex-critical (P_5, P_5) -free graphs for fixed k [10]. Hell and Huang proved that there are finitely many k -vertex-critical (P_6, C_4) -free graphs [14]. This was later generalized to $(P_t, K_{r,s})$ -free graphs in the context of H -coloring [21]. This gives an affirmative answer for $H = K_{2,3}$. In [3], Cai, Goedgebeur and Huang showed that there are finitely many k -vertex-critical (P_5, gem) -free graphs and finitely many k -vertex-critical $(P_5, P_3 + P_2)$ -free graphs. Later, Cameron and Hoàng [6] gave a better bound on the order of k -vertex-critical (P_5, gem) -free graphs and determined all such graphs for $k \leq 7$. Moreover, it has been proven that there are finitely many 5-vertex-critical (P_5, bull) -free graphs [19] and finitely many 5-vertex-critical (P_5, chair) -free graphs [17]. We refer the interested reader to [7] for more details.

Our contributions. A dart (see Fig. 1) is the graph obtained from a diamond by adding a new vertex and making it adjacent to exactly one vertex with degree 3 in the diamond. Our main result is as follows.

Theorem 4. *For every fixed integer $k \geq 1$, there are finitely many k -vertex-critical (P_5, dart) -free graphs.*

We prove a Ramsey-type statement (see Lemma 4) which allows us to prove our main result by induction on k . This is another example of a result that is proved by induction on k for the finiteness problem besides the one in [10].

We perform a careful structural analysis via Strong Perfect Graph Theorem combined with the pigeonhole principle based on the properties of vertex-critical graphs. Moreover, for $k \in \{5, 6, 7\}$ we computationally determine a list of all k -vertex-critical (P_5, dart) -free graphs.

Our results imply the existence of polynomial-time certifying algorithm to decide the k -colorability of (P_5, dart) -free graphs for $k \geq 1$. (An algorithm is *certifying* if, along with the answer given by the algorithm, it also gives a certificate which allows to verify in polynomial time that the output of the algorithm is indeed correct; in case of k -coloring the certificate is either a k -coloring or a $(k + 1)$ -vertex-critical induced subgraph.)

Theorem 5. *For every fixed integer $k \geq 1$, there is a polynomial-time certifying algorithm to decide the k -colorability of (P_5, dart) -free graphs.*

Proof. Let G be a (P_5, dart) -free graph. We first run the polynomial-time algorithm for determining whether a P_5 -free graph is k -colorable from [15] for G . If the answer is yes, the algorithm outputs a k -coloring of G . Otherwise G is not k -colorable. In this case, G must contain a $(k + 1)$ -vertex-critical (P_5, dart) -free graph as an induced subgraph. For each $(k + 1)$ -vertex-critical (P_5, dart) -free graph H , it takes polynomial-time to check whether G contains H . Since there are only finitely many such graphs by Theorem 4, we can do this for every such graph H and the total running time is still polynomial. $\square$

The remainder of the paper is organized as follows. We present some preliminaries in Section 2 and give structural properties around an induced C_5 in a (P_5, dart) -free graph in Section 3. We show that there are finitely many k -vertex-critical (P_5, dart) -free graphs for all $k \geq 1$ in Section 4 and computationally determine an exhaustive list of such graphs for $k \in \{5, 6, 7\}$ in Section 5. Finally, we give a conclusion in Section 6.

2. Preliminaries

For general graph theory notation we follow [1]. For $k \geq 4$, an induced cycle of length k is called a k -hole. A k -hole is an *odd hole* (respectively *even hole*) if k is odd (respectively even). A k -antihole is the complement of a k -hole. Odd and even antiholes are defined analogously.

Let $G = (V, E)$ be a graph. If $uv \in E(G)$, we say that u and v are *neighbors* or *adjacent*, otherwise u and v are *nonneighbors* or *nonadjacent*. The *neighborhood* of a vertex v , denoted by $N_G(v)$, is the set of neighbors of v . For a set $X \subseteq V(G)$, let $N_G(X) = \bigcup_{v \in X} N_G(v) \setminus X$. We shall omit the subscript whenever the context is clear. For $x \in V(G)$ and $S \subseteq V(G)$, we denote by $N_S(x)$ the set of neighbors of x that are in S , i.e., $N_S(x) = N_G(x) \cap S$. For two sets $X, S \subseteq V(G)$, let $N_S(X) = \bigcup_{v \in X} N_S(v) \setminus X$.

For $X, Y \subseteq V(G)$, we say that X is *complete* (resp. *anticomplete*) to Y if every vertex in X is adjacent (resp. nonadjacent) to every vertex in Y . If $X = \{x\}$, we write “ x is complete (resp. anticomplete) to Y ” instead of “ $\{x\}$ is complete (resp. anticomplete) to Y ”. If a vertex v is neither complete nor anticomplete to a set S , we say that v is *mixed* on S . For a vertex $v \in V$ and an edge $xy \in E$, if v is mixed on $\{x, y\}$, we say that v is *mixed* on xy . For a set $H \subseteq V(G)$, if no vertex in $V(G) \setminus H$ is mixed on H , we say that H is a *homogeneous set*, otherwise H is a *nonhomogeneous set*.

A vertex subset $S \subseteq V(G)$ is *stable* if no two vertices in S are adjacent. A *clique* is the complement of a stable set. Two nonadjacent vertices u and v are said to be *comparable* if $N(v) \subseteq N(u)$ or $N(u) \subseteq N(v)$. For an induced subgraph A of G , we write $G - A$ instead of $G - V(A)$. For $S \subseteq V$, the subgraph induced by S is denoted by $G[S]$. We say that a vertex w *distinguishes* two vertices u and v if w is adjacent to exactly one of u and v .

We proceed with a few useful results that will be used later. The first folklore property of vertex-critical graph is that such graphs contain no comparable vertices. A generalization of this property was presented in [5].

Lemma 1 ([5]). *Let G be a k -vertex-critical graph. Then G has no two nonempty disjoint subsets X and Y of $V(G)$ that satisfy all the following conditions.*

- X and Y are anticomplete to each other.
- $\chi(G[X]) \leq \chi(G[Y])$.
- Y is complete to $N(X)$.

Lemma 2 ([19]). *Let G be a 5-vertex-critical P_5 -free graph and S be a homogeneous set of $V(G)$. For each component A of $G[S]$,*

- (i) *if $\chi(A) = 1$, then A is a K_1 ;*
- (ii) *if $\chi(A) = 2$, then A is a K_2 ;*
- (iii) *if $\chi(A) = 3$, then A is a K_3 or a C_5 .*

We extend Lemma 2 to all critical graphs, which may be of independent interest.

Lemma 3. *Let G be a k -vertex-critical graph and S be a homogeneous set of $V(G)$. For each component A of $G[S]$, if $\chi(A) = m$ with $m < k$, then A is an m -vertex-critical graph.*

Proof. Suppose not. Then there exist some vertex $u \in V(A)$ such that $\chi(A - u) = m$. Since G is k -vertex-critical, $G - u$ is $(k - 1)$ -colorable. Let ϕ be a $(k - 1)$ -coloring of $G - u$ and $\phi(A - u) = \{1, 2, \dots, s\}$ with $m \leq s \leq k - 1$. Since A is homogeneous, then $\phi(N(A - u)) \notin \{1, 2, \dots, s\}$. Let f be an m -coloring of A and $f(A) = \{1, 2, \dots, m\}$. Let $g(A) = f$ and $g(G - A) = \phi|_{G-A}$. Then g is a $(k - 1)$ -coloring of G . This contradicts G is k -vertex-critical.

Therefore, A is an m -vertex-critical graph. $\square$

Next, we prove an important lemma, which will be used frequently in the proof of our results.

Lemma 4. *Let G be a dart-free graph and c be a fixed integer. Let S, T be two disjoint subsets of $V(G)$ such that $|S| \leq c$ and $\overline{G}[T \cup S]$ is connected. If every vertex in S is adjacent to at least one of w_1 and w_2 where w_1 and w_2 are two arbitrary nonadjacent vertices in T , and there exists a vertex $w \in V(G) \setminus (S \cup T)$ such that w is complete to $S \cup T$, then $|T| \leq c(\chi(T))^1 + c(\chi(T))^2 + \dots + c(\chi(T))^{2\chi(T)+1}$.*

Proof. Let $N_0 = S$, $N_i = \{v | v \in T \setminus \bigcup_{j=0}^{i-1} N_j, v \text{ has a nonneighbor in } N_{i-1}\}$ where $i \geq 1$. Let $u \in N_{i-1}$ and $v, v' \in N_i$ with $vv' \notin E(G)$. Since $\overline{G}[T \cup S]$ is connected, $N_i \neq \emptyset$.

First, we show that u is adjacent to at least one of v and v' . If $i = 1$, we are done. Now we consider the cases of $i \geq 2$. Suppose that u is adjacent to neither v nor v' . Let $u' \in N_{i-2}$ be the nonneighbor of u . Then u' is complete to v and v' by the definition of N_i . Since w is complete to $S \cup T$. Then $\{v, v', u', w, u\}$ induces a dart. So u is adjacent to at least one of v and v' .

Next, we show that $|N_i| \leq c(\chi(T))^i$. Since $\chi(N_i) \leq \chi(T)$, we can partition N_i into at most $\chi(T)$ stable sets. Because u is adjacent to at least one of v and v' , each vertex in N_{i-1} has at most $\chi(T)$ nonneighbors in N_i . Since $|N_0| = |S| \leq c$, we have $|N_i| \leq c(\chi(T))^i$.

Finally, we show that $i \leq 2\chi(T) + 1$. Suppose not. By the definition of N_i , N_i and N_j are complete when $|i - j| > 1$. Then we take a vertex $u_i \in N_i$ where i is even. Now, $\{u_2, u_4, \dots, u_{2\chi(T)+2}\}$ induces a $K_{\chi(T)+1}$, a contradiction. This shows $i \leq 2\chi(T) + 1$.

Therefore, $|T| \leq c(\chi(T))^1 + c(\chi(T))^2 + \dots + c(\chi(T))^{2\chi(T)+1}$. $\square$

The following theorem tells us there are finitely many 4-vertex-critical P_5 -free graphs.

Theorem 6 ([2,23]). *If $G = (V, E)$ is a 4-vertex-critical P_5 -free graph, then $|V| \leq 13$.*

A property on bipartite graphs is shown as follows.

Lemma 5 ([11]). *Let G be a connected bipartite graph. If G contains a $2K_2$, then G must contain a P_5 .*

The clique number of G , denoted by $\omega(G)$, is the size of a largest clique in G . A graph G is *perfect* if $\chi(H) = \omega(H)$ for every induced subgraph H of G . Another result we use is the famous Strong Perfect Graph Theorem.

Theorem 7 (The Strong Perfect Graph Theorem [8]). *A graph is perfect if and only if it contains no odd holes or odd antiholes.*

3. Structure around 5-hole

Let $G = (V, E)$ be a graph and H be an induced subgraph of G . We partition $V \setminus V(H)$ into subsets with respect to H as follows: for any $X \subseteq V(H)$, we denote by $S(X)$ the set of vertices in $V \setminus V(H)$ that have X as their neighborhood among $V(H)$, i.e.,

$$S(X) = \{v \in V \setminus V(H) : N_{V(H)}(v) = X\}.$$

For $0 \leq m \leq |V(H)|$, we denote by S_m the set of vertices in $V \setminus V(H)$ that have exactly m neighbors in $V(H)$. Note that $S_m = \bigcup_{X \subseteq V(H): |X|=m} S(X)$.

Let G be a (P_5, dart) -free graph and $C = v_1, v_2, v_3, v_4, v_5$ be an induced C_5 in G . We partition $V \setminus C$ with respect to C as follows, where all indices below are modulo five.

$$\begin{aligned} S_0 &= \{v \in V \setminus V(C) : N_C(v) = \emptyset\}, \\ S_2(i) &= \{v \in V \setminus V(C) : N_C(v) = \{v_{i-1}, v_{i+1}\}\}, \\ S_3^1(i) &= \{v \in V \setminus V(C) : N_C(v) = \{v_{i-1}, v_i, v_{i+1}\}\}, \\ S_3^2(i) &= \{v \in V \setminus V(C) : N_C(v) = \{v_{i-2}, v_i, v_{i+2}\}\}, \\ S_4(i) &= \{v \in V \setminus V(C) : N_C(v) = \{v_{i-2}, v_{i-1}, v_{i+1}, v_{i+2}\}\}, \\ S_5 &= \{v \in V \setminus V(C) : N_C(v) = V(C)\}. \end{aligned}$$

Let $S_3^1 = \bigcup_{i=1}^5 S_3^1(i)$, $S_3^2 = \bigcup_{i=1}^5 S_3^2(i)$ and $S_4 = \bigcup_{i=1}^5 S_4(i)$. Since G is P_5 -free, we have that

$$V(G) = S_0 \cup S_2 \cup S_3^1 \cup S_3^2 \cup S_4 \cup S_5.$$

We now prove a number of useful properties of these sets using the fact that G is (P_5, dart) -free. All properties are proved for $i = 1$ due to symmetry.

- (1) S_0 is anticomplete to $S_2 \cup S_3^1$.
Let $u \in S_0$, $v \in S_2(1) \cup S_3^1(1)$. If $uv \in E(G)$, then $\{u, v, v_2, v_3, v_4\}$ is an induced P_5 .
- (2) For each $1 \leq i \leq 5$, $S_3^2(i)$ is not mixed on any edge of S_0 .
Let uu' be an edge of S_0 , and $v \in S_3^2(1)$. If $uv \in E(G)$ and $uv' \notin E(G)$, then $\{u', u, v, v_1, v_2\}$ is an induced P_5 .
- (3) S_0 is anticomplete to $S_4 \cup S_5$.
Let $u \in S_0$ and $v \in S_4(1) \cup S_5$. If $uv \in E(G)$, then $\{v_3, v_4, v_5, v, u\}$ is a dart.
- (4) Let A be a component of S_0 , then A is homogeneous and P_3 -free.
By (1)–(3), we have A is homogeneous. Suppose that A contains a $P_3 = uvw$. Since G is connected, there must exist some vertex $x \in N(A)$. By (1)–(3), $x \in S_3^1(i)$ for some $1 \leq i \leq 5$. Then $\{u, v, w, x, v_i\}$ is a dart. So, A is P_3 -free.
- (5) For each $1 \leq i \leq 5$, $S_3^1(i)$ is not mixed on any edge of $S_2(i)$.
Let uu' be an edge of $S_2(1)$ and $v \in S_3^1(1)$. If $vu' \in E(G)$ and $vu \notin E(G)$, then $\{u, u', v, v_2, v_3\}$ is a dart.
- (6) For each $1 \leq i \leq 5$, $S_2(i)$ is complete to $S_3^1(i+1) \cup S_3^1(i-1)$.
Let $u \in S_2(1)$, $v \in S_3^1(2)$. If $uv \notin E(G)$, then $\{v_1, v, v_3, v_2, u\}$ is a dart. By symmetry, $S_2(1)$ is complete to $S_3^1(5)$.
- (7) For each $1 \leq i \leq 5$, $S_2(i)$ is anticomplete to $S_3^1(i+2) \cup S_3^1(i-2)$.
Let $u \in S_2(1)$, $v \in S_3^1(3)$. If $uv \in E(G)$, then $\{v_1, v_5, u, v, v_3\}$ is an induced P_5 . By symmetry, $S_2(1)$ is anticomplete to $S_3^1(4)$.
- (8) For each $1 \leq i \leq 5$, $S_2(i)$ is complete to $S_3^2(i)$.
Let $u \in S_2(1)$ and $v \in S_3^2(1)$. If $uv \notin E(G)$, then $\{u, v_5, v_1, v, v_3\}$ is an induced P_5 .

- (9) For each $1 \leq i \leq 5$, $S_2(i)$ is anticomplete to $S_3^2 \setminus S_3^2(i)$.
 Let $u \in S_2(1)$, $v \in S_3^2(2)$. If $uv \in E(G)$, then $\{u, v, v_4, v_5, v_1\}$ is a dart. By symmetry, $S_2(1)$ is anticomplete to $S_3^1(5)$.
 Let $w \in S_3^2(3)$. If $uw \in E(G)$, then $\{u, w, v_1, v_5, v_3\}$ is a dart. By symmetry, $S_2(1)$ is anticomplete to $S_3^1(4)$.
- (10) For each $1 \leq i \leq 5$, $S_2(i)$ is anticomplete to $S_4(i)$.
 Let $u \in S_2(1)$, $v \in S_4(1)$. If $uv \in E(G)$, then $\{u, v, v_3, v_2, v_1\}$ is a dart.
- (11) For each $1 \leq i \leq 5$, $S_2(i)$ is complete to $S_4 \setminus S_4(i)$.
 Let $u \in S_2(1)$, $v \in S_4(2) \cup S_4(3)$. If $uv \notin E(G)$, then $\{v_1, v, v_4, v_5, u\}$ is a dart. By symmetry, $S_2(1)$ is complete to $S_4(5) \cup S_4(4)$.
- (12) For each $1 \leq i \leq 5$, $S_2(i)$ is complete to $S_2(i+1) \cup S_2(i-1)$.
 Let $u \in S_2(1)$, $v \in S_2(2)$. If $uv \notin E(G)$, then $\{u, v_5, v_1, v, v_3\}$ is an induced P_5 . By symmetry, $S_2(1)$ is complete to $S_2(5)$.
- (13) For each $1 \leq i \leq 5$, $S_2(i+2) \cup S_2(i-2)$ is not mixed on any edge of $S_2(i)$.
 Let uu' be an edge of $S_2(1)$ and $v \in S_2(3)$. If $vu' \in E(G)$ and $vu \notin E(G)$, then $\{v_3, v_4, v, u', u\}$ is an induced P_5 . By symmetry, $S_2(4)$ is not mixed on any edge of $S_2(1)$.
- (14) If $S_2 \neq \emptyset$, then $S_5 = \emptyset$. If $S_5 \neq \emptyset$, then $S_2 = \emptyset$.
 Suppose that $S_2 \neq \emptyset$ and $S_5 \neq \emptyset$. Let $u \in S_2(1)$ and $v \in S_5$. If $uv \notin E(G)$, then $\{v_1, v, v_4, v_5, u\}$ is a dart. If $uv \in E(G)$, then $\{u, v_2, v_1, v, v_4\}$ is a dart.
- (15) Let A be a component of $S_2(i)$. Then A is homogeneous and A is P_3 -free.
 By (5)–(14), we have A is a homogeneous set. Suppose that A contains a $P_3 = uvw$, then $\{u, v, w, v_{i-1}, v_i\}$ is a dart. So A is P_3 -free.
- (16) For each $1 \leq i \leq 5$, S_5 is complete to $S_3^2(i)$.
 Let $u \in S_5$, $v \in S_3^2(1)$. If $uv \notin E(G)$, then $\{v_2, u, v_5, v_1, v\}$ is a dart.
- (17) Let $u, u' \in S_5$ with $uu' \notin E(G)$. Then every vertex in $S_3^1 \cup S_4$ is adjacent to at least one of u and u' .
 Let $v \in S_3^1(1) \cup S_4(4)$. If v is adjacent to neither u nor u' , then $\{u, u', v_4, v_5, v\}$ is a dart.
- (18) For $1 \leq i \leq 5$, $S_3^1(i)$, $S_3^2(i)$ and $S_4(i)$ is a clique, respectively.
 Let $u, v \in S_3^1(1)$. If $uv \notin E(G)$, then $\{u, v, v_1, v_5, v_4\}$ is a dart. So $S_3^1(1)$ is a clique. Let $u, v \in S_3^2(1)$. If $uv \notin E(G)$, then $\{u, v, v_4, v_3, v_2\}$ is a dart. So $S_3^2(1)$ is a clique. Let $u, v \in S_4(1)$. If $uv \notin E(G)$, then $\{u, v, v_3, v_2, v_1\}$ is a dart. So $S_4(1)$ is a clique.

4. The proof of Theorem 4

Proof. We prove the theorem by induction on k . If $1 \leq k \leq 4$, there are finitely many k -vertex-critical (P_5 , dart)-free graphs by Theorem 6. In the following, we assume that $k \geq 5$ and there are finitely many i -vertex-critical graphs for $i \leq k-1$. Now, we consider the cases of k .

Let $G = (V, E)$ be a k -vertex-critical (P_5 , dart)-free graph. We show that G has finite order. Let $\mathcal{L} = \{K_k, \overline{C_{2k-1}}\}$. If G has a subgraph isomorphic to a member $L \in \mathcal{L}$, then $|V(G)| = |V(L)|$ by the definition of vertex-critical and so we are done. So, we assume in the following G has no induced subgraph isomorphic to a member in $\mathcal{L}$. Then G is imperfect. Since G is k -vertex-critical and $\chi(\overline{C_{2t+1}}) \geq k+1$ if $t \geq k$, it follows that G does not contain $\overline{C_{2t+1}}$ for $t \geq k$. Moreover, since G is P_5 -free, it does not contain $\overline{C_{2t+1}}$ for $t \geq 3$. It then follows from Theorem 7, G must contain some $\overline{C_{2t+1}}$ for $2 \leq t \leq k-2$. To finish the proof, we only need to prove the following two lemmas.

4.1. 5-hole

Lemma 6. If G contains an induced C_5 , then G has finite order.

Proof. Let $C = v_1, v_2, v_3, v_4, v_5$ be an induced C_5 . We partition $V(G)$ with respect to C .

Since G is K_k -free, combined with (18), we have $S_3^1(i)$, $S_3^2(i)$ and $S_4(i)$ is K_{k-2} -free, respectively. Then $|S_3^1(i)| \leq k-3$, $|S_3^2(i)| \leq k-3$ and $|S_4(i)| \leq k-3$. Therefore, $|S_3^1| \leq 5k-15$, $|S_3^2| \leq 5k-15$ and $|S_4| \leq 5k-15$. Thus, in the following, we only need to bound S_0 , S_2 and S_5 .

We first bound S_0 .

Claim 1. Let A be a component of S_0 , then $\chi(A) \leq k-2$.

Proof. Suppose that $\chi(A) \geq k-1$. Since G is connected, there must exist $v \in N(A)$ such that $v \in S_3^2$ by (1)–(3). Since A is homogeneous by (4), v is complete to A . Then $\chi(G[V(A) \cup \{v\}]) \geq k$. Since $G[V(A) \cup \{v\}] \subset G$, it contradicts with G is k -vertex-critical. $\square$

Let A be a component of G , we call A a K_i -component if $A \cong K_i$ where $i \geq 1$.

Claim 2. Each component of S_0 is a K_m where $1 \leq m \leq k-2$. For each $1 \leq m \leq k-2$, the number of K_m -components is not more than 2^{5k-15} .

Proof. By (4), S_0 is a disjoint union of cliques. Combined with Claim 1, it follows that each component of S_0 is a K_m where $1 \leq m \leq k-2$.

Suppose that the number of K_1 -components in S_0 is more than $2^{5k-15} \geq 2^{|S_3^1|}$. The pigeonhole principle shows that there are two K_1 -components u, v having the same neighborhood in S_3^1 . Since S_0 is anticomplete to $S_2 \cup S_3^1 \cup S_4 \cup S_5$. Then u, v have the same neighborhood in $V(G)$. This contradicts with Lemma 1.

Similarly, we can show that the number of K_2 -components, K_3 -components, ..., K_{k-2} -components is not more than 2^{5k-15} , respectively. $\square$

By Claim 2, it follows that S_0 is bounded. Next, we bound S_2 and S_5 . Note that at least one of S_2 and S_5 is an empty set by (14). In the following, we first assume that $S_5 \neq \emptyset$. Then $S_2 = \emptyset$.

Claim 3. S_5 is bounded.

Proof. Let $N_0 = S_3^1 \cup S_4$, $N_i = \{v | v \in S_5 \setminus \bigcup_{j=0}^{i-1} N_j, v \text{ has a nonneighbor in } N_{i-1}\}$ where $i \geq 1$.

Assume that $G[S_5 \cup S_3^1 \cup S_4]$ is connected. Since $|S_3^1 \cup S_4| \leq 10k-30$ and there exists a $w \in V(C)$ such that w is complete to $S_3^1 \cup S_4 \cup S_5$. By (17), every vertex in $S_3^1 \cup S_4$ is adjacent to at least one of w_1 and w_2 where $w_1, w_2 \in S_5$ and $w_1 w_2 \notin E(G)$. Then $|S_5|$ is a function of $\chi(S_5)$ by Lemma 4. Since $\chi(S_5) \leq k-3$, it follows that S_5 is bounded.

Assume that $G[S_5 \cup S_3^1 \cup S_4]$ is not connected. Then there exists an integer $j \geq 0$ such that $N_0, N_1, \dots, N_j \neq \emptyset$ but $N_{j+1} = \emptyset$. Then $S_5 - \bigcup_{i=0}^j N_i$ is complete to $\bigcup_{i=0}^j N_i$, and so $S_5 - \bigcup_{i=0}^j N_i$ is a homogeneous set. Since $\chi(S_5 - \bigcup_{i=0}^j N_i) \leq k-3$, each component of $S_5 - \bigcup_{i=0}^j N_i$ is an m -vertex-critical graph with $1 \leq m \leq k-3$ by Lemma 3. By the inductive hypothesis, it follows that there are finitely many m -vertex-critical (P_5, dart) -free graphs with $1 \leq m \leq k-3$. By the pigeonhole principle, the number of each kind of graph is not more than 2. So, $S_5 - \bigcup_{i=0}^j N_i$ is bounded. For each N_i with $1 \leq i \leq j$, N_i is bounded by Lemma 4. Therefore, S_5 is bounded. $\square$

Thus, G has finite order if $S_5 \neq \emptyset$. Next, we assume that $S_2 \neq \emptyset$. Then $S_5 = \emptyset$ by (14).

Claim 4. Each component of $S_2(i)$ is a K_m where $1 \leq m \leq k-2$. For each $1 \leq m \leq k-2$, the number of K_m -components is not more than $2^{k-2} + 1$.

Proof. By (15), $S_2(i)$ is a disjoint union of cliques. If $S_2(i)$ contains a K_{k-1} , then G contains a K_k , a contradiction. So each component of $S_2(i)$ is a K_m where $1 \leq m \leq k-2$.

Suppose that the number of K_1 -components in $G[S_2(1)]$ is more than $2^{k-2} + 1 \geq 2 \cdot 2^{|S_3^1(1)|} + 1$. By the pigeonhole principle, there are y_1, y_2, y_3 having the same neighborhood in $S_3^1(1)$. For any $y_i \neq y_j$ with $i, j \in \{1, 2, 3\}$, since y_i and y_j are not comparable, there must exist $y'_i \in N(y_i) \setminus N(y_j)$, $y'_j \in N(y_j) \setminus N(y_i)$. By (5)–(13), $y'_i, y'_j \in S_2(3) \cup S_2(4)$.

If $y'_i, y'_j \in S_2(3)$, then $y'_i y'_j \in E(G)$. For otherwise $\{y'_i, y_i, v_5, y_j, y'_j\}$ is an induced P_5 . Then $\{y'_i, y'_j, y_j, v_2, v_1\}$ is a dart, a contradiction. By symmetry, $y'_i, y'_j \notin S_2(4)$. In the following, by symmetry, we assume that $y'_i \in S_2(3)$, $y'_j \in S_2(4)$. Then $y'_i y'_j \in E(G)$, otherwise $\{y_j, y'_j, v_3, v_4, y'_i\}$ is an induced P_5 . Let $y_l \neq y_i, y_j$ with $1 \leq l \leq 3$. Then $y_l y'_i \notin E(G)$, otherwise $\{y_i, y'_i, y_l, v_2, v_1\}$ is a dart. By symmetry, $y_l y'_j \notin E(G)$. Since y_i and y_l are not comparable, there must exist $y'_l \in N(y_l) \setminus N(y_i)$. Then $y'_l \in S_2(4)$, $y_l y'_l \notin E(G)$ and $y_l y'_j \notin E(G)$. Then $y'_l y'_j \in E(G)$, otherwise $\{y_l, y'_l, v_3, y'_j, y_j\}$ is an induced P_5 . Then $\{y_j, y'_j, y'_l, v_5, v_1\}$ is a dart, a contradiction.

Similarly, we can show that the number of K_2 -components, K_3 -components, ..., K_{k-2} -components is not more than $2^{k-2} + 1$, respectively. $\square$

Therefore, G has finite order if $S_2 \neq \emptyset$. This completes the proof of Lemma 6. $\square$ $\square$

4.2. $(2t+1)$ -antihole

Lemma 7. If G contains an induced $\overline{C_{2t+1}}$ for $3 \leq t \leq k-2$, then G has finite order.

Proof. Let $C = v_1, v_2, \dots, v_{2t+1}$ be an induced $\overline{C_{2t+1}}$ such that $v_i v_j \in E(G)$ if and only if $|i-j| > 1$. All indices are modulo $2t+1$. We partition $V(G)$ with respect to C . For any $X \subseteq V(C)$ with $|X| = m$, we denote by $S_m(X) = \{v \in V(G) \setminus V(C) : N_C(v) = X\}$. In the following, we shall write $S_2(v_i, v_{i+1})$ for $S_2(\{v_i, v_{i+1}\})$, $S_3(v_i, v_{i+1}, v_{i+3})$ for $S_3(\{v_i, v_{i+1}, v_{i+3}\})$, etc.

Claim 5. $S_m = \emptyset$ for $1 \leq m \leq t$.

Proof. Let $u \in S_m$ for some $1 \leq m \leq t$. Since u has at most t neighbors on C , there exists an index i such that u is adjacent to neither v_{i-t} nor v_{i+t} . Note that $v_{i+t} v_{i-t} \notin E(G)$. Let $U = V(C) \setminus \{v_{i-t}, v_{i-t+1}, v_{i+t}, v_{i+t-1}\}$.

If u is mixed on U , then there exist $j, s \in \{i-t+2, i-t+3, \dots, i, i+1, \dots, i+t-2\}$ with $|j-s| \geq 2$ such that u distinguishes v_j and v_s . Then $\{v_{i-t}, v_{i+t}, v_j, v_s, u\}$ induces a dart.

If u is complete to U , then u is anticomplete to $\{v_{i-t+1}, v_{i+t-1}\}$. For otherwise u would be complete to $\{v_{i-t+1}, v_{i-t+2}, \dots, v_{i+t-1}\}$ which contradicts the assumption that u has at most t neighbors on C . Then $\{v_{i-t+1}, v_{i+t}, v_{i+t-1}, v_i, u\}$ induces a dart.

Now assume that u is anticomplete to U . By symmetry, we assume that u is adjacent to v_{i-t+1} . Then $\{u, v_{i-t+1}, v_{i+t}, v_{i-t+2}, v_{i-t}\}$ is an induced P_5 .

Therefore, $S_m = \emptyset$, for $1 \leq m \leq t$. $\square$

Claim 6. $S_0 = \emptyset$.

Proof. Suppose not. Let $u \in S_0$. By Claim 5 and the connectivity of G , u has a neighbor $u' \in S_i$ for some $t+1 \leq i \leq 2t+1$. Then there exists an index i such that u' is adjacent to v_{i-t}, v_{i+t} and some vertex v_j where $j \in \{i-t+2, \dots, i, i+1, \dots, i+t-2\}$. Now $\{v_{i-t}, v_{i+t}, v_j, u', u\}$ induces a dart. This proves $S_0 = \emptyset$. $\square$

Claim 7. For every $X \subseteq C$ with $t+1 \leq |X| \leq 2t$, $S(X)$ is a clique.

Proof. Let $X \subseteq C$ be an arbitrary set with $t+1 \leq |X| \leq 2t$. Since $t+1 \leq |X| \leq 2t$, there exists an index $i \in C$ such that $v_i \in X$ but $v_{i+1} \notin X$, and $v_j \in X$ where $v_j \in V(C) \setminus \{v_{i-2}, v_{i-1}, \dots, v_{i+2}\}$. If $S(X)$ contains two non-adjacent vertices u and u' , then $\{u, u', v_i, v_j, v_{i+1}\}$ induces a dart. Therefore, $S(X)$ is a clique. $\square$

Since G is K_k -free, by Claim 7, it follows that $|S(X)| \leq k-3$ for $X \subseteq C$ with $t+1 \leq |X| \leq 2t$. Therefore, $|S_{t+1} \cup S_{t+2} \cup \dots \cup S_{2t}| \leq (k-3)\binom{2t+1}{t+1} + \dots + \binom{2t+1}{2t}$. In the following, we bound S_{2t+1} .

Claim 8. Let $u, u' \in S_{2t+1}$ with $uu' \notin E(G)$. Then every vertex in $S_{t+1} \cup \dots \cup S_{2t}$ is adjacent to at least one of u and u' .

Proof. Let $v \in S_m$ with $t+1 \leq m \leq 2t$. Then there exist two vertices $v_i, v_j \in V(C)$ with $|i-j| \geq 2$ such that v distinguishes v_i and v_j . If v is adjacent to neither u nor u' , then $\{u, u', v_i, v_j, v\}$ induces a dart. $\square$

Claim 9. S_{2t+1} is bounded.

Proof. Let $N_0 = S_{t+1} \cup S_{t+2} \cup \dots \cup S_{2t}$, $N_i = \{v | v \in S_{2t+1} \setminus \bigcup_{j=0}^{i-1} N_j, v \text{ has a nonneighbor in } N_{i-1}\}$ where $i \geq 1$.

Assume that $\overline{G[S_{t+1} \cup S_{t+2} \cup \dots \cup S_{2t+1}]}$ is connected. Since $|S_{t+1} \cup S_{t+2} \cup \dots \cup S_{2t}| \leq (k-3)\binom{2t+1}{t+1} + \dots + \binom{2t+1}{2t}$ and there exists $w \in V(C)$ such that w is complete to $S_{t+1} \cup S_{t+2} \cup \dots \cup S_{2t+1}$. By Claim 9, every vertex in $S_{t+1} \cup S_{t+2} \cup \dots \cup S_{2t}$ is adjacent to at least one of w_1 and w_2 where $w_1, w_2 \in S_{2t+1}$ and $w_1 w_2 \notin E(G)$. Then $|S_{2t+1}|$ is a function of $\chi(S_{2t+1})$ by Lemma 4. Since $\chi(S_{2t+1}) \leq k-t-1$, it follows that S_{2t+1} is bounded.

Assume that $\overline{G[S_{t+1} \cup S_{t+2} \cup \dots \cup S_{2t+1}]}$ is not connected. Then there exists an integer $j \geq 0$ such that $N_1, N_2, \dots, N_j \neq \emptyset$ but $N_{j+1} = \emptyset$. Then $S_{2t+1} - \bigcup_{i=0}^j N_i$ is complete to $\bigcup_{i=0}^j N_i$, and so $S_{2t+1} - \bigcup_{i=0}^j N_i$ is a homogeneous set. Since $\chi(S_{2t+1} - \bigcup_{i=0}^j N_i) \leq k-t-1$, each component of $S_{2t+1} - \bigcup_{i=0}^j N_i$ is an m -vertex-critical graph with $1 \leq m \leq k-4$ by Lemma 3. By the inductive hypothesis, it follows that there are finitely many m -vertex-critical (P_5, dart) -free graph for $1 \leq m \leq k-4$. By the pigeonhole principle, the number of each kind of graph is not more than $2^{|S_{t+1} \cup \dots \cup S_{2t}|}$. So, $S_{2t+1} - \bigcup_{i=0}^j N_i$ is bounded. For each N_i with $1 \leq i \leq j$, N_i is bounded by Lemma 4. Therefore, S_{2t+1} is bounded. $\square$

This completes the proof of Lemma 7. $\square$

By Lemma 6–Lemma 7, it follows that Theorem 4 holds. $\square$

5. Complete characterization for $k \in \{5, 6, 7\}$

In Section 4, we proved that there are finitely many k -vertex-critical (P_5, dart) -free graphs by showing the existence of an upper bound for the order of such graphs for every integer $k \geq 1$. These bounds are not necessarily sharp. In the current section, we show sharp upper bounds for $k \in \{5, 6, 7\}$ by computationally determining an exhaustive list of all k -vertex-critical (P_5, dart) -free graphs.

We created two independent implementations of the algorithm from Goedgebeur and Schaudt [13], which we extended for the (P_5, dart) -free case. The source code of these implementations can be downloaded from [12,20]. The algorithm expects three parameters as an input: an integer $k \geq 1$, a set of graphs $\mathcal{H}$ and a graph I . It generates all k -vertex-critical $\mathcal{H}$ -free graphs that contain I as an induced subgraph. The pseudocode is given in Algorithm 1. The algorithm is not guaranteed to terminate (e.g. there could be infinitely many such graphs). However, if the algorithm terminates, it is guaranteed that the generated graphs are exhaustive. The algorithm works by recursively extending a graph with one vertex and adding edges between this new vertex and already existing vertices. In each step of the recursion, the algorithm uses powerful pruning rules that restrict the ways in which the edges are added. These pruning rules allow the algorithm to terminate some branches of the recursion and are the reason why the algorithm itself can terminate in some cases. For example, a graph which is not $(k-1)$ -colorable cannot appear as a proper induced subgraph of a

Table 1The number of k -critical (P_5, dart) -free and k -vertex-critical (P_5, dart) -free graphs for $k \in \{5, 6, 7\}$.

Vertices	5	6	7	8	9	10	11	Total
5-critical	1		1	1	7	1		
5-vertex-critical	1		1	6	172	1		
6-critical		1		1	1	6	33	
6-vertex-critical		1		1	6	171	17,834	
7-critical			1		1	1	6	
7-vertex-critical			1		1	6	171	
Vertices	12	13	14	15	16	19		
5-critical		3						14
5-vertex-critical		3						184
6-critical	2	1			13			58
6-vertex-critical	2	1			13			18,029
7-critical	28	250	6	2	1	35		331
7-vertex-critical	17,834	6,349,644	6	2	1	35		6,367,701

k -vertex-critical graph, so the algorithm does not need to extend such graphs. In general, the pruning rules that are used by the algorithm are much more sophisticated than this. We refer the interested reader to [13] for further details about the correctness of the algorithm and the different pruning rules.

Algorithm 1 Extend(An integer k , A set of graphs $\mathcal{H}$, A graph I)

```

1: if  $I$  is  $\mathcal{H}$ -free AND not generated before then
2:   if  $I$  is not  $(k - 1)$ -colorable then
3:     if  $I$  is a  $k$ -vertex-critical graph then
4:       output  $I$ 
5:     end if
6:   else
7:     for every graph  $I'$  obtained by adding a new vertex  $u$  to  $I$  and edges between  $u$  and vertices in  $V(I)$  in all possible
       ways that are permitted by the pruning rules do
8:       Extend( $k, \mathcal{H}, I'$ )
9:     end for
10:  end if
11: end if

```

We now prove the following characterization theorem:

Theorem 8. *There are exactly 184 5-vertex-critical (P_5, dart) -free graphs and the largest such graphs have order 13. There are exactly 18,029 6-vertex-critical (P_5, dart) -free graphs and the largest such graphs have order 16. There are exactly 6,367,701 7-vertex-critical (P_5, dart) -free graphs and the largest such graphs have order 19.*

Proof. We saw in Section 4 that every k -vertex-critical (P_5, dart) -free graph is either $K_k, \overline{C_{2k-1}}$ or contains C_5 as an induced subgraph or contains $\overline{C_{2t+1}}$ as an induced subgraph for some $2 \leq t \leq k - 2$. If Algorithm 1 is called with the parameters $k \in \{5, 6, 7\}$, $\mathcal{H} = \{P_5, \text{dart}\}$ and I one of these graphs, the algorithm terminates in less than a second for $k = 5$, less than a minute for $k = 6$ and a few hours for $k = 7$ (for all choices of k and I). The counts of these graphs are reported in Table 1. The results of the two independent implementations of this algorithm (cf. [12,20]) are in complete agreement with each other. $\square$

Table 1 gives an overview of the number of k -critical and k -vertex-critical (P_5, dart) -free graphs for $k \in \{5, 6, 7\}$. A graph G is k -critical (P_5, dart) -free if it is k -chromatic, (P_5, dart) -free and every (P_5, dart) -free proper subgraph of G is $(k - 1)$ -colorable. The graphs from Table 1 can be obtained from the meta-directory of the *House of Graphs* [9] at <https://houseofgraphs.org/meta-directory/critical-h-free>. Moreover, the k -critical graphs from Table 1 can also be inspected in the searchable database of the *House of Graphs* [9] by searching for the keywords “critical (P_5, dart) -free”. The 5-critical (P_5, dart) -free graphs are shown in Fig. 2. We also tried characterizing the 8-vertex-critical (P_5, dart) -free graphs, but the algorithm did not terminate after running for two weeks.

6. Conclusion

In this paper, we have proved that there are finitely many k -vertex-critical (P_5, dart) -free graphs for $k \geq 1$ and computationally determined an exhaustive list of such graphs for $k \in \{5, 6, 7\}$. Our results gave an affirmative answer to

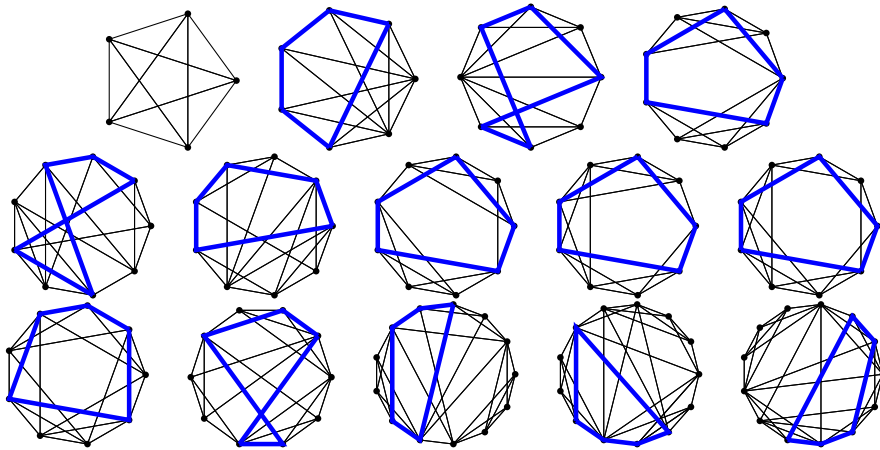


Fig. 2. All 14 5-critical (P_5, dart) -free graphs. The blue edges indicate an induced C_5 . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the problem posed in [5] for $H = \text{dart}$. In the future, it is natural to investigate the finiteness of the set of k -vertex-critical (P_5, H) -free graphs for other graphs H of order 5. In fact, it only leaves the cases of $H \in \{P_4 + P_1, \text{diamond} + P_1, C_4 + P_1, P_3 + 2P_1, W_4\}$ for $k \geq 5$, $H \in \{\text{chair}, \text{bull}, K_5\}$ for $k \geq 6$ and $H = K_5 - e$ for $k = 5, 6, 7$ as open problem.

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Data availability

No data was used for the research described in the article.

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